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The Most Profound Discovery of Science

John and Mary Bell arrived to a nation in mourning. The day before they landed in California, President Kennedy had been shot and killed in Dallas. “It was the worst possible moment to have come,” John said later. John and Mary, both specialists in the physics of particle accelerators, had been invited to spend a year as academic guests at the Stanford Linear Accelerator Center (SLAC), half a world away from their usual professional home in Switzerland. Despite the tragedy, they set to work. “Mary was quickly integrated into the accelerator division,” John recalled, “and I into the [particle] theory group.”

John used the change of scenery as an opportunity to explore scientific ideas that had been weighing on his mind for over a decade. Ever since reading David Bohm’s papers in 1952, Bell had known there was something wrong with von Neumann’s famous proof that purportedly showed theories like Bohm’s pilot-wave interpretation couldn’t work. Yet other physicists still regularly cited von Neumann as justification for ignoring Bohm’s ideas. Shortly before Bell left Switzerland, he had spoken with Josef Jauch, a physicist at the University of Geneva who had recently published a “strengthened” version of von Neumann’s proof. Jauch defended his own ideas and pointed Bell toward another proof that also supposedly ruled out Bohm’s version of quantum physics. “For me, that was like a red light to a bull,” Bell said. “I wanted to show that Jauch was wrong. We had gotten